Where to Hitch? Finding Small, Rare Parts Across Long Spatiotemporal Contexts

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Abstract

Successful hitch assistance requires precisely locating a trailer's coupler so that the vehicle's hitch ball can connect to it. However, this small socket is uncommon in everyday images, making its name an unreliable cue for vision foundation models. Text prompts often lead these models to identify a larger surrounding structure instead. We introduce **HitchIt**, a framework for localizing the coupler throughout a recorded video without further model training. HitchIt draws on three insights. First, suitable visual examples specify the intended target more reliably than text prompts. Second, hierarchical search uses known part relationships to focus detection on relevant image regions. Third, tracking uses temporal consistency to propagate a reliable identification from an informative view throughout the video, avoiding an independent search in every frame. With SAM3, HitchIt achieves 0.627 mean box intersection over union on 20 videos evaluated frame by frame. Text-prompted models, including SAM3 with a text prompt, reach at most 0.067. Ablations show the benefits of example selection and hierarchical search. We release dense coupler annotations through the HitchIt annotation source.

1 Introduction

Hitch assistance must align a vehicle's hitch ball with a trailer's coupler to connect them [1, 2]. The coupler is the socket at the tip of the trailer's tongue, the frame extending from its body, as shown in Fig. 1. Successful alignment requires precisely locating the coupler as the vehicle backs up.

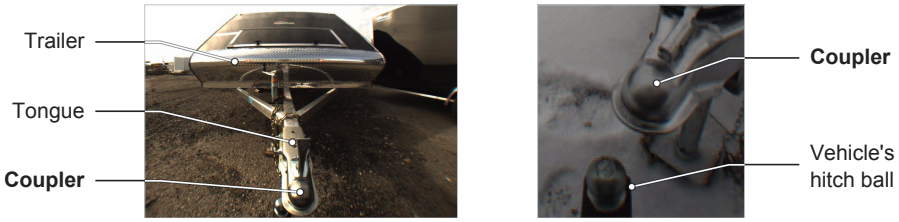


Figure 1: **Hitch assistance connects the hitch ball to the coupler.** The left scene shows the trailer–tongue–coupler structure; the right scene distinguishes the coupler from the vehicle's hitch ball.

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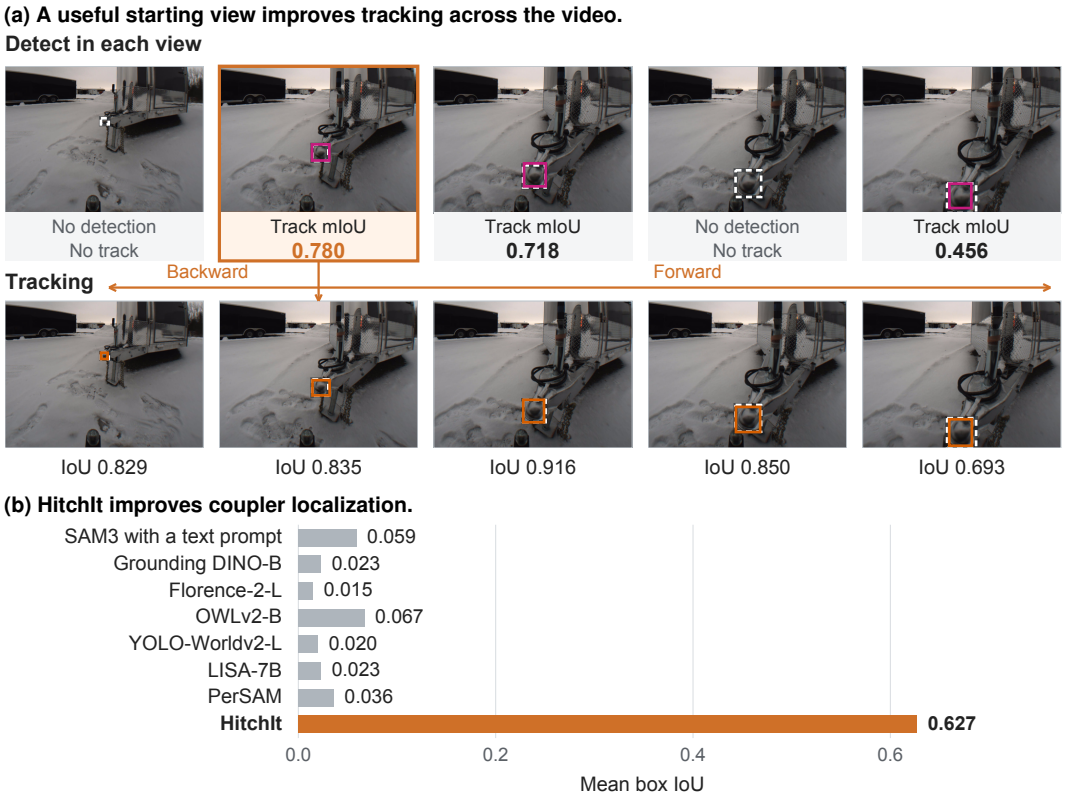


Figure 2: **Identify the coupler in a useful view, then track across the video. (a)** The starting view affects the resulting track. Top: the same HitchIt detector searches each view; its box starts the same tracker. Labels report whole-video mIoU. Bottom: the highlighted starting view yields accurate tracks even where detection fails. Orange arrows connect this view to tracking in both directions. Magenta: detections; orange: tracked boxes; dashed white: annotations. IoU is box intersection over union; mIoU averages it over annotated frames. **(b)** HitchIt outperforms existing methods [3–9] on 20 videos. SAM3 searches four late views with a text prompt, then tracks its most confident box.

Vision foundation models achieve strong performance in object detection and semantic localization [3–5]. However, small, uncommon parts remain difficult to localize. The coupler resembles other metal parts, and its rarity in everyday images makes its name an unreliable cue. In our evaluation, text-prompted models, including SAM3 with a text prompt, often identify larger surrounding structures.

We introduce HitchIt, a framework that locates couplers throughout recorded videos without further training. It draws on three insights. First, suitable visual examples specify the intended part more reliably than text prompts. Second, hierarchical search uses known part relationships to focus detection on relevant image regions. Third, tracking uses temporal consistency to propagate a reliable identification from an informative view across the video, as shown in Figures 2a and 3.

HitchIt searches from the trailer to its tongue and then to the coupler, selecting matching examples from a varied reference bank at each stage. Cropping before resizing preserves detail. HitchIt selects a coupler detection from the searched views and tracks it backward and forward through the video.

On 20 videos evaluated frame by frame, HitchIt outperforms text-prompted models, including SAM3 with a text prompt, as shown in Fig. 2b. Controlled comparisons show the benefits of example selection, hierarchical search, and view selection. Evaluation on 67 additional videos, each scored at one labeled frame, further supports hierarchical search over full-image search.

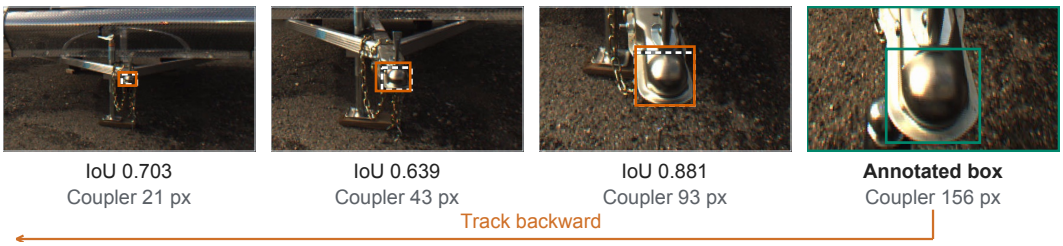


Figure 3: **A correct starting box enables tracking across changes in target size.** The green annotated box in the rightmost view starts tracking backward; orange boxes are predictions and dashed white boxes are annotations. This isolates tracking from automatic detection. Close-ups share one display scale; coupler size is $\sqrt{\text{box area}}$ in original pixels.

Our contributions are as follows. (1) **Localization guided by examples and structure:** HitchIt combines suitable examples, hierarchical grounding, and tracking to locate a small, rare part without further training or a target annotation in the new video. (2) **Controlled localization studies:** we measure how example selection, hierarchical search, and view selection affect identification and the resulting tracks while holding the search or tracker fixed. (3) **Dense coupler annotations:** we release coupler boxes throughout 20 videos from the MSU Trailer Database [1] for future research.

2 Related Work

Long videos and tracking. Long-video benchmarks test whether models can use information spread across a recording [10–12]. Video segmentation methods follow a marked target by retaining information from earlier frames [5, 13–15]. HitchIt uses a pretrained tracker unchanged and focuses on finding a reliable starting box in a suitable view. Selective labeling chooses samples for human annotation [16, 17]; HitchIt chooses frames for automatic localization.

Text and visual prompts. Detectors and segmenters can use text to specify objects beyond a fixed list of categories [3, 4, 6–8]. Visual prompts specify the intended target with examples [5, 18–20]. Other methods segment new images from annotated examples using pretrained features without further training [9, 21, 22]. HitchIt matches reference parts to the search region, then presents each selected example beside that region with its target marked by a box.

Guided visual search. V* [23] uses a model that processes images and language to search high-resolution images by zooming into likely regions. HitchIt grounds search in known part relationships, following the trailer to its tongue and then the coupler at its tip.

Coupler perception. Prior hitch-assist systems train detectors on task-specific labels [1, 2]. The MSU Trailer Database provides approach videos with coupler centers and close-up contours [1]. We release dense coupler boxes for 20 videos and study localization without further model training.

3 HitchIt

Task. HitchIt receives a recorded approach video and fixed reference banks with boxed examples of the trailer, tongue, and coupler. It predicts coupler boxes without an annotation in the new video. Using the full recording lets an informative view guide tracking in both directions.

Overview. HitchIt specifies the target with suitable visual examples, grounds hierarchical search in known part relationships, and uses temporal consistency to track from a selected detection. The pretrained model must support detection from a marked example and tracking from a box; its weights remain unchanged. Figure 4 shows how the examples, regions, and views guide localization.

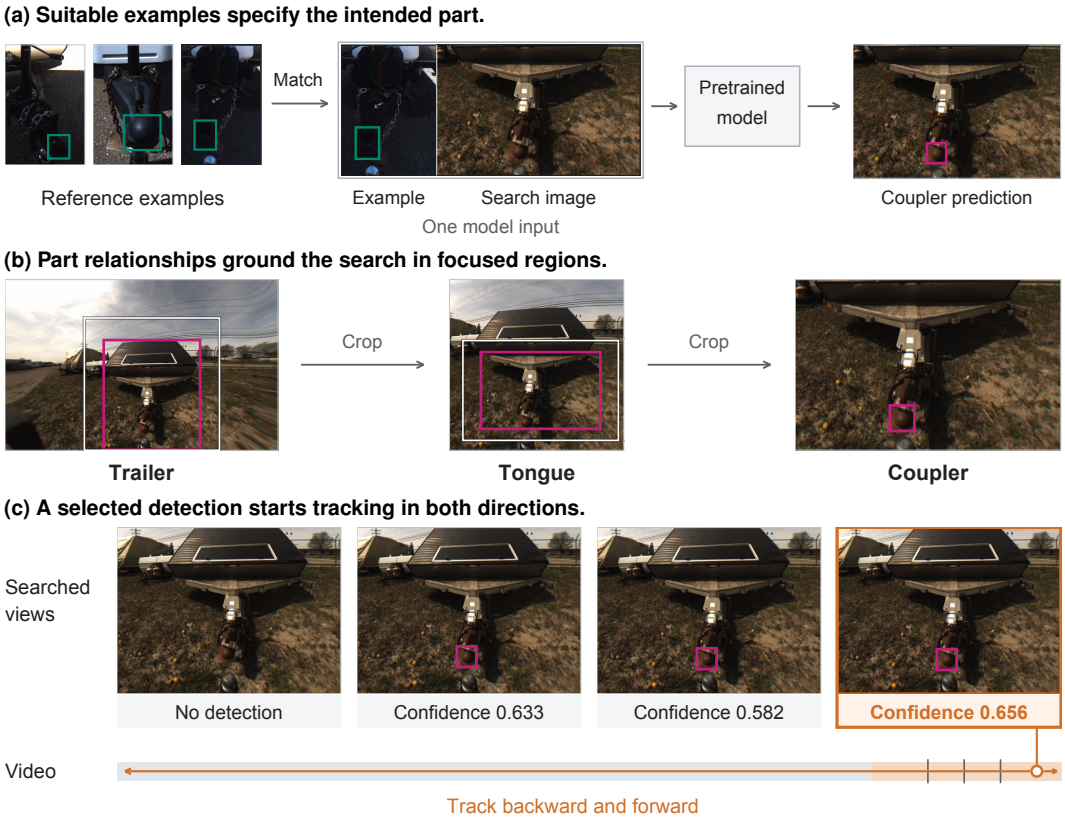


Figure 4: **HitchIt guides localization with examples, structure, and temporal consistency.** **(a)** A matching reference example and the search image form one model input; the green box marks the intended part in the example. **(b)** The trailer–tongue–coupler hierarchy focuses each detection on the relevant region, using this example pairing at every stage. White outlines mark the next search region. **(c)** Four frames from the shaded final fifth are searched. The orange connector links the selected detection to its position in the video, where tracking starts toward both ends. Scores are detection confidence. Magenta: detections; green: annotated examples; orange: tracking.

3.1 Specify the target with suitable visual examples

A fixed reference bank provides examples of each part with varied appearances, viewpoints, and backgrounds. The examples specify which part to find; boxes mark that part within each example.

Match examples by part appearance. HitchIt ranks examples by how closely the marked part resembles content in the search image. It averages the model's image features inside each example box and compares this vector with the features at each search-image location. The score averages the eight highest cosine similarities, and HitchIt keeps the eight best-scoring examples. This *part matching* ranks references using the marked part rather than the whole image.

Present each example beside the search image. HitchIt presents each selected example beside the search image as one model input, with a box marking only the reference part, as shown in Fig. 4a. For each of the eight pairs, it keeps the most confident prediction on the search-image side. It removes overlapping duplicate detections by keeping higher-confidence boxes, a step called non-maximum suppression, then ranks the remaining boxes by confidence.

3.2 Ground hierarchical search in part relationships

The tongue extends from the trailer body, with the coupler at its tip. These relationships ground the coupler in the trailer's structure and focus detection. In a full-image pair, the search image occupies half of the model's fixed input, leaving few pixels for the coupler.

HitchIt first finds the trailer in the full image, then the tongue in a crop around each trailer, and finally the coupler in a crop around the tongue. Figure 4b shows this *hierarchical search*. Each stage pairs its region with examples of that part; coupler examples mark the target within a tongue crop. Cropping before resizing enlarges the next part and excludes surrounding objects.

HitchIt keeps up to three candidate trailers and rejects tongues and couplers with implausible size or position. For example, a tongue must lie in the lower part of its trailer box, and a coupler box may cover at most a quarter of its tongue box's area. When the tongue crop yields no valid coupler, HitchIt searches the full image for a coupler inside the detected tongue, and then searches inside the trailer.

3.3 Identify in a useful view, then track

Temporal consistency lets tracking propagate an identification from a suitable view across the recording. Searching N frames with all R reference examples takes NR model calls per part. With 99 examples per bank, this would require $4{,}851$ to $17{,}424$ calls per part per video. HitchIt searches four evenly spaced frames in the last fifth, when the coupler generally appears larger, and selects a starting detection from these candidates. Figure 4c illustrates the search and tracking. Among valid coupler boxes, HitchIt prefers those found through a detected tongue over those found only in the trailer. It then selects the most confident box, breaking near-ties toward the larger box, and slightly enlarges the box if it is small. HitchIt tracks the selected box backward and forward to both ends of the video. Without a valid coupler detection in the searched frames, it produces no track.

4 Experiments

We compare HitchIt with existing methods and examine the contributions of example selection, hierarchical search, and view selection. We also evaluate more videos at one labeled frame each.

4.1 Experimental setup

Data. We use rear-camera approach videos from the MSU Trailer Database [1]. The frame-by-frame evaluation uses 20 videos with 49–176 sampled 1920×1200 frames each, $1{,}784$ frames in total. We annotated coupler boxes on $1{,}136$ of these frames and score all methods on them. The remaining frames, all at the start of a video, carry no annotation and are not scored. The coupler covers a median 0.2% of the image. We measure its size as $\sqrt{\text{box area}}$ in pixels; within a video, this varies by a median factor of 6.6 between the 10th and 90th percentiles. For this frame-by-frame evaluation, each fixed reference bank holds 99 boxed examples from 99 videos outside this evaluation set. HitchIt has one bank per part, and its coupler examples are crops around the tongue.

Metrics. Box intersection over union, denoted IoU, divides intersection area by union area: 0 means no overlap and 1 an exact match. Mean IoU, abbreviated mIoU, averages the IoU of the best-matching predicted box over annotated frames, counting a missing prediction as zero. AP_{50} and AR_{50} are COCO average precision and recall [24] at IoU 0.5; recall allows one box per frame. We adapt cumulative IoU, abbreviated cIoU, from text-guided segmentation [8] to boxes. It divides the total intersection by the total union over annotated frames, including unmatched predicted boxes in the denominator, so oversized and extra boxes lower it. mIoU is our main metric because it credits partial overlap without relying on confidence scores, which are constant within each track.

Implementation. We evaluate HitchIt with SAM3 [5], using its visual-example prompts and video tracker unchanged. Part matching uses SAM3's pretrained Perception Encoder features [25]. SAM3 resizes each input to 1008×1008, so in a full-image pair the search image receives 504 pixels of input width. Weights stay fixed; each predicted mask becomes its smallest enclosing box.

Compared methods. We use "trailer coupler" as the default text prompt. Grounding DINO-B [3], Florence-2-L [6], OWLv2-B [4], YOLO-Worldv2-L [7], and LISA-7B [8] process every frame independently and return one box per frame. SAM3 receives this text prompt either on the first video frame or separately on four late views; the most confident late-view detection starts its tracker.

Table 1: **HitchIt localizes the coupler more accurately than existing methods.** Results on the 1,136 annotated frames of the 20 videos. Tracked rows search the listed views, then track. Text*: the better of one or three target names. Bold: best automatic result. The annotated-box row measures tracking from a supplied correct box rather than automatic localization.

Method	Prompt	Searched views	mIoU	AP_{50}	AR_{50}	cIoU
Text-prompted foundation models						
SAM3 [5], first frame	text	first frame → track	0.038	0.000	0.000	0.106
SAM3 [5], late views	text	4 late views → track	0.059	0.000	0.000	0.099
Grounding DINO-B [3]	text	every frame	0.023	0.000	0.000	0.022
Florence-2-L [6]	text	every frame	0.015	0.000	0.000	0.016
OWLv2-B [4]	text*	every frame	0.067	0.000	0.002	0.097
YOLO-Worldv2-L [7]	text*	every frame	0.020	0.000	0.000	0.013
LISA-7B [8]	text	every frame	0.023	0.000	0.000	0.020
Example-prompted methods: no box in the new video						
PerSAM [9]	boxed examples	4 late views → track	0.036	0.000	0.003	0.014
HitchIt	boxed examples	4 late views → track	**0.627**	**0.564**	**0.750**	**0.632**
Tracking from an annotated box supplied in the test video						
Annotated box	test-video box	1 frame → track	0.683	0.616	0.787	0.759

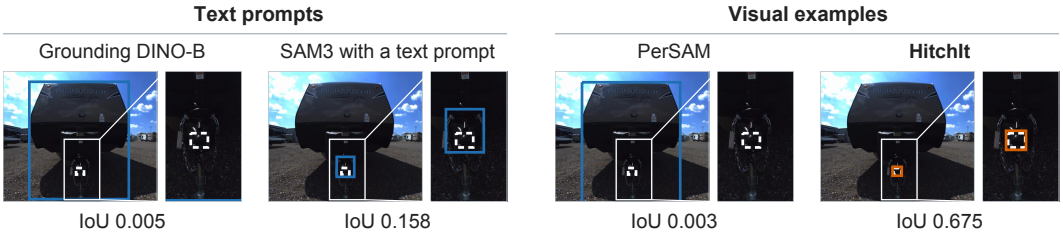


Figure 5: **Finding a larger structure does not identify the coupler.** Grounding DINO-B [3], late-view SAM3 text prompting [5], PerSAM [9], and HitchIt on the same frame, each with a full view and a close-up of the outlined region. PerSAM follows Table 1's boxed-example and tracking protocol; its box extends below the full view. Blue: baseline predictions; orange: HitchIt; dashed white: the annotated coupler box. Only HitchIt isolates the coupler in this example.

PerSAM [9] receives eight full-image coupler examples per view, retrieved by part matching, with their boxes converted to masks. It uses HitchIt's four late views and tracker. To evaluate tracking from a known target, we start the tracker from each video's last annotated box, as in Fig. 3.

Direct search baseline. *Direct search* skips the trailer and tongue stages. It retrieves full-image coupler examples by part matching and searches the full image, with HitchIt's views and tracker.

4.2 Comparison with existing methods

HitchIt localizes the coupler more accurately than text prompting. In Table 1, HitchIt reaches 0.627 mIoU. Text-prompted settings reach 0.015 to 0.067, all with an AP_{50} of 0.000. Even on the same four late views, SAM3 reaches only 0.059. Size alone does not explain the gap: text prompts reach at most 0.108 mIoU on 517 frames with couplers at least 80 pixels, versus 0.715 for HitchIt.

Text and example prompts can identify the wrong structure. Figure 5 shows Grounding DINO-B and PerSAM locating the trailer instead of the coupler. Late-view SAM3 text prompting locates the hitch assembly, while HitchIt identifies the coupler. PerSAM reaches 0.036 mIoU despite receiving boxed coupler examples. On 92.5% of frames, Grounding DINO-B returns a box that covers at least half of the coupler but is at least four times its area; its median box is 63 times the coupler's area.

Matching size is not enough to identify the coupler. OWLv2 proposes a box with IoU ≥ 0.5 on 752 of the 1,136 frames, but its text score selects such a box on only two.

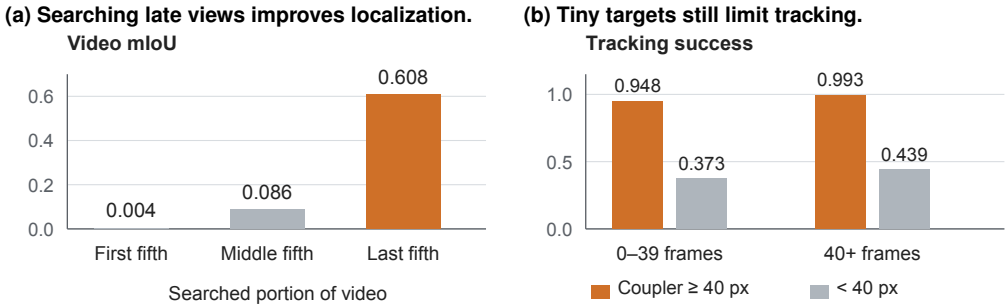


Figure 6: **View selection supports identification; small size still limits tracking. (a)** Direct search on four views from the first, middle, or last fifth of each video, followed by the same tracker. **(b)** With an annotated starting box, tracking success is the fraction of frames with box IoU ≥ 0.5. Results are grouped by distance from that view in sampled frames and by coupler size.

A detailed description does not help either. With the prompt "small metal socket at the tip of the trailer tongue", Grounding DINO-B's median predicted area shrinks to 1.6 times the coupler's area, but only two frames reach IoU ≥ 0.5. Its errors include a frame fitting, a decal, and the vehicle's hitch ball, so matching the coupler's scale does not single it out.

4.3 Identify in an informative view before tracking

View selection improves identification and tracking. To isolate the choice of views, we fix direct search and the tracker and change only where the four searched views come from: the first, middle, or last fifth of each video. If none of the four views yields a box, no track is produced. In Fig. 6a, mIoU rises from 0.004 for the first fifth and 0.086 for the middle fifth to 0.608 for the last fifth. Late views show a larger coupler, while the choice among these views also affects identification and tracking. In the controlled example in Fig. 2a, tracking from the highlighted detection gives 0.780 mIoU, versus 0.456 when the same tracker starts from the final view.

A correct starting box enables effective tracking. Figure 3 illustrates how one correct box lets the tracker follow the coupler through earlier views. In Table 1, tracking from the annotated starting box gives 0.683 mIoU across all 20 videos, whereas the name prompt on the late views gives 0.059. The starting box matters: HitchIt reaches 0.627 by selecting its box automatically.

Tiny couplers remain hard to track. Figure 6b groups tracking success from the annotated starting box by coupler size and distance from that frame. Success requires box IoU of at least 0.5. For couplers at least 40 pixels in size, success is 0.948 within 39 sampled frames of the starting view and 0.993 beyond. For smaller couplers, these rates drop to 0.373 and 0.439, respectively.

4.4 Select suitable examples and ground the search

Table 2 varies how examples are retrieved and where the coupler is searched, with the four late views and the tracker fixed. *Scene matching* ranks examples by the similarity of feature vectors describing the whole image, from CLIP [26] or from SAM3's encoder, instead of part matching. Two spatial baselines use HitchIt's cropped coupler examples without trailer or tongue predictions. One searches the full image; the other searches six overlapping tiles of half its width and two-thirds its height.

Part appearance improves reference selection. In Table 2, part matching gives 0.608 mIoU in direct search, compared with 0.590 for scene matching with CLIP and 0.544 with SAM3 features.

Hierarchical grounding improves the search region. In Table 2, with the same cropped coupler examples, hierarchical search reaches 0.627 mIoU, versus 0.460 for six fixed tiles and 0.486 for full-image search. The tiles require six times as many model calls as full-image search. Hierarchical search also improves mIoU, AP_{50}, and AR_{50} over direct search with full-image examples.

Table 2: **Part matching and hierarchical search improve localization.** All rows use the same four late frames of the 20 videos and the same tracker. The first group varies example retrieval; the last two compare search regions with cropped coupler examples. Bold: best.

Search setup	mIoU	AP_{50}	AR_{50}	cIoU
Direct search: how examples are retrieved				
Scene matching, CLIP [26]	0.590	0.456	0.676	0.587
Scene matching, SAM3 features	0.544	0.413	0.634	0.542
Part matching, SAM3 features	0.608	0.512	0.705	**0.657**
Direct search with cropped coupler examples				
Full image	0.486	0.442	0.607	0.485
Six fixed tiles	0.460	0.360	0.555	0.465
Hierarchical search				
HitchIt: trailer $\rightarrow$ tongue $\rightarrow$ coupler	**0.627**	**0.564**	**0.750**	0.632

Table 3: **Hierarchical search improves on direct search across separate videos.** Results on 67 videos, each scored on one labeled late frame using the tracked box. Every test video is excluded from its example bank. Both methods use part matching and the same tracker. This tests cross-video localization, not tracking accuracy across all frames.

Search setup	mIoU	AP_{50}	AR_{50}	cIoU
Direct search	0.513	0.349	0.537	0.438
HitchIt, hierarchical search	**0.645**	**0.718**	**0.761**	**0.649**

4.5 Evaluation on a wider range of videos

We evaluate HitchIt on 67 additional videos from the MSU Trailer Database. Each has one annotated coupler box on a frame 90 to 93% through the video. We split the videos into five groups and process each group with example banks built from the other four groups, which gives 53 or 54 examples per part. We use each test video's annotation only to score its tracked box.

Hierarchical search improves localization across videos. In Table 3, hierarchical search improves all four metrics over direct search. It raises mIoU from 0.513 to 0.645 and roughly doubles AP_{50}, from 0.349 to 0.718. All five video groups show positive mIoU gains, ranging from 0.003 to 0.244.

5 Conclusion

Hitch assistance depends on locating the coupler, the small socket that receives the vehicle's hitch ball. We introduced HitchIt to localize this uncommon part throughout a recorded video without further model training. Suitable visual examples specify the intended target, hierarchical search uses known part relationships to guide detection, and tracking uses temporal consistency to propagate a reliable identification from an informative view throughout the video.

On 20 videos evaluated frame by frame, HitchIt reaches a mean box overlap of 0.627, versus at most 0.067 for automatic baselines, including SAM3 with a text prompt and PerSAM. On 67 additional videos scored at one labeled frame each, hierarchical search improves mean overlap from 0.513 to 0.645 over full-image search. The comparisons support selecting informative examples, regions, and views before tracking through the recording. Very small couplers remain difficult to track, even with a correct starting box. The approach may extend to other uncommon parts with similar-looking neighbors and known part relationships. We release dense coupler annotations through the HitchIt annotation source to support further research on small-part localization in video.

Acknowledgments

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References

[1] Yousef Atoum, Joseph Roth, Michael Bliss, Wende Zhang, and Xiaoming Liu. Monocular video-based trailer coupler detection using multiplexer convolutional neural network. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 5477–5485, 2017.

[2] Ashok Dahal, Jakir Hossen, Sumanth Chennupati, Ganesh Sistu, Kazumi Malhan, Muhammad Amasha, and Senthil Yogamani. DeepTrailerAssist: Deep learning based trailer detection, tracking and articulation angle estimation on automotive rear-view camera. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pages 2339–2346, 2019.

[3] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, 2024.

[4] Matthias Minderer, Alexey Gritsenko, and Neil Houlsby. Scaling open-vocabulary object detection. In *Advances in Neural Information Processing Systems*, 2023.

[5] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris Coll-Vinent, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu, Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollar, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. SAM 3: Segment anything with concepts. In *The Fourteenth International Conference on Learning Representations*, 2026. URL `https://openreview.net/forum?id=r35clVtGzw`.

[6] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[7] Tianheng Cheng, Lin Song, Yixiao Ge, Wenyu Liu, Xinggang Wang, and Ying Shan. YOLO-World: Real-time open-vocabulary object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[8] Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, Yuhui Yuan, Shu Liu, and Jiaya Jia. LISA: Reasoning segmentation via large language model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[9] Renrui Zhang, Zhengkai Jiang, Ziyu Guo, Shilin Yan, Junting Pan, Hao Dong, Yu Qiao, Peng Gao, and Hongsheng Li. Personalize segment anything model with one shot. In *International Conference on Learning Representations (ICLR)*, 2024.

[10] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, et al. Ego4D: Around the world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[11] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-MME: The first-ever comprehensive evaluation benchmark of multi-modal LLMs in video analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[12] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. LongVideoBench: A benchmark for long-context interleaved video-language understanding. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2024.

[13] Seoung Wug Oh, Joon-Young Lee, Ning Xu, and Seon Joo Kim. Video object segmentation using space-time memory networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019. URL `https://openaccess.thecvf.com/content_ICCV_2019/papers/Oh_Video_Object_Segmentation_Using_Space-Time_Memory_Networks_ICCV_2019_paper.pdf`.

[14] Ho Kei Cheng and Alexander G. Schwing. XMem: Long-term video object segmentation with an Atkinson-Shiffrin memory model. In *European Conference on Computer Vision*, 2022.

[15] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/45c1f6a8cbf2da59ebf2c802b4f742cd-Abstract-Conference.html.

[16] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations*, 2018. URL https://openreview.net/forum?id=H1aIuk-RW.

[17] Xudong Wang, Long Lian, and Stella X. Yu. Unsupervised selective labeling for more effective semi-supervised learning. In *European Conference on Computer Vision*, 2022. URL https://www.ecva.net/papers/eccv_2022/papers_ECCV/html/2575_ECCV_2022_paper.php.

[18] Qing Jiang, Feng Li, Zhaoyang Zeng, Tianhe Ren, Shilong Liu, and Lei Zhang. T-Rex2: Towards generic object detection via text-visual prompt synergy. In *European Conference on Computer Vision (ECCV)*, pages 38–57, 2024. doi: 10.1007/978-3-031-73414-4_3.

[19] Feng Li, Qing Jiang, Hao Zhang, Tianhe Ren, Shilong Liu, Xueyan Zou, Huaizhe Xu, Hongyang Li, Jianwei Yang, Chunyuan Li, Lei Zhang, and Jianfeng Gao. Visual in-context prompting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12861–12871, 2024.

[20] Tianhe Ren, Yihao Chen, Qing Jiang, Zhaoyang Zeng, Yuda Xiong, Wenlong Liu, Zhengyu Ma, Junyi Shen, Yuan Gao, Xiaoke Jiang, Xingyu Chen, Zhuheng Song, Yuhong Zhang, Hongjie Huang, Han Gao, Shilong Liu, Hao Zhang, Feng Li, Kent Yu, and Lei Zhang. DINO-X: A unified vision model for open-world object detection and understanding. arXiv preprint arXiv:2411.14347, 2024. URL https://arxiv.org/abs/2411.14347.

[21] Yang Liu, Muzhi Zhu, Hengtao Li, Hao Chen, Xinlong Wang, and Chunhua Shen. Matcher: Segment anything with one shot using all-purpose feature matching. In *International Conference on Learning Representations (ICLR)*, 2024.

[22] Miguel Espinosa, Chenhongyi Yang, Linus Ericsson, Steven McDonagh, and Elliot J. Crowley. No time to train! Training-free reference-based instance segmentation. arXiv preprint arXiv:2507.02798v4, 2026. URL https://arxiv.org/abs/2507.02798v4.

[23] Penghao Wu and Saining Xie. V*: Guided visual search as a core mechanism in multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13084–13094, 2024.

[24] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*, pages 740–755, 2014. doi: 10.1007/978-3-319-10602-1_48.

[25] Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception Encoder: The best visual embeddings are not at the output of the network. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-2036.

[26] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.